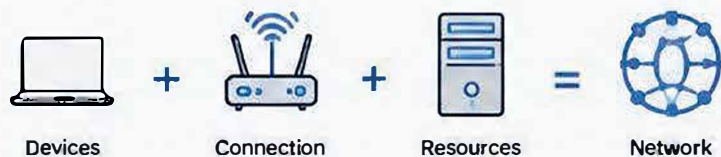


Networking Fundamentals

The foundation of how devices connect, communicate, and share resources.

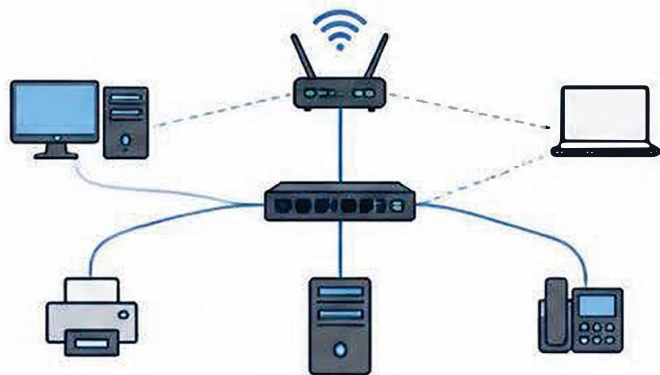
1. WHAT IS A NETWORK?

A network is a group of two or more devices connected together so they can communicate, share data, and access resources.



3. HOW IT WORKS

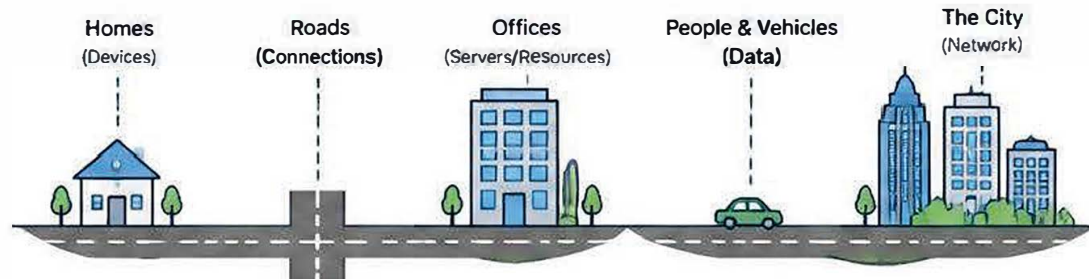
Devices connect using cables or Wi-Fi. They communicate by sending data to each other.



Data travels across the connection, following rules so the right device receives it.

2. REAL-WORLD ANALOGY: A CITY

Think of a network like a city.

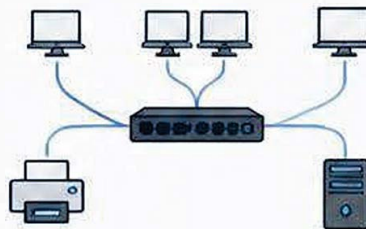


Roads connect buildings so people and vehicles can travel and share.
Networks connect devices so data can travel and be shared.

4. LAN vs WAN

LAN (Local Area Network)

Connects devices within a small area, like a home, office, or school.



Example: Office network

WAN (Wide Area Network)

Connects LANs over large distances, using the internet or other networks.



Example: Company offices in different cities

5. WHY IT MATTERS



Share resources
Printers, files, applications, and internet.



Improve communication
Email, chat, video calls, and collaboration.



Increase efficiency
Access data and tools anytime, anywhere.



Enhance reliability & security
Backups, permission controls, and secure access.



What is a Server OS?

Understanding Server OS vs. Client OS



Definition

A **Server OS** (Operating System) is software that manages hardware resources and provides services, data, and applications to other computers (called clients) over a network.

Its job is to be reliable, secure, and available so that many users can connect and get the resources they need.

	Client OS (Personal Computer)	Server OS (Always-On, Multi-User)
Primary Purpose	Help one person get work done (productivity, entertainment)	Provide services and resources to many users and systems
Users	Typically 1 user at a time	Many users simultaneously
Uptime	Can be turned off or restarted as needed	Designed to run 24/7 with high availability
Resource Role	Consumes resources	Manages and distributes resources (CPU, memory, storage, network)
Focus	Ease of use and personal experience	Reliability, security, scalability, and centralized management
Examples	Windows 11, Windows 10, macOS	Windows Server, Linux Server Distributions

Real-World Analogy

Client OS

like a personal car



Built for one person or a small group.
Goes where you want, when you want.

VS.

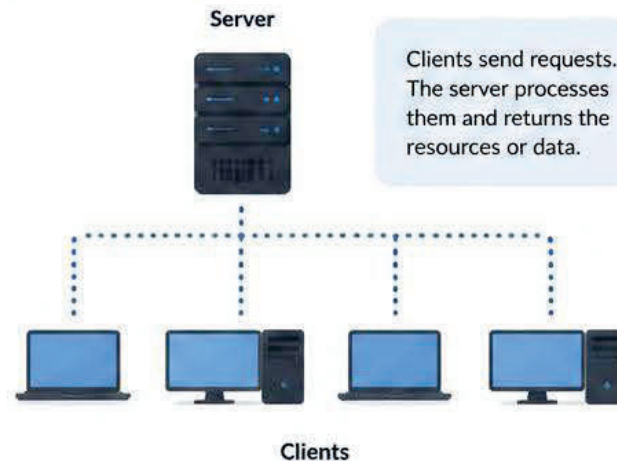
Server OS

like a taxi dispatch center



Always on. Coordinates many drivers (clients),
assigns jobs, shares resources, and keeps
everything running smoothly.

How It Works



Why It Matters



Centralized data and apps mean everyone works with the same accurate information.



Stronger security and backup protect important data.



Servers handle more users and data as your organization grows.



IT can manage, update, and monitor everything from one place.



Always-on services keep your business running.



Windows Server Roles & Features

Understanding the Building Blocks of a Windows Server



What is a ROLE?

A Role is a major server function. It provides core services that the server is designed to deliver.

Examples:

- File and Storage Services (file sharing)
- Print and Document Services (printing)
- Active Directory Domain Services (authentication)



What is a FEATURE?

A Feature is a supporting tool or capability that enhances a Role. It adds functionality but does not deliver a standalone core service.

Examples:

- .NET Framework (supports applications)
- BitLocker Drive Encryption (secures data)
- PowerShell (automation and management)

Aspect	ROLE	FEATURE
Purpose	Provides a major server service or function.	Provides additional tools or capabilities that support roles.
Scope	Broad, core functionality.	Narrower, supplemental functionality.
Can it run alone?	Yes, a role delivers value by itself.	No, a feature enhances a role but is not useful on its own.
Examples	• File and Storage Services • Print and Document Services • Active Directory Domain Services	• .NET Framework • BitLocker Drive Encryption • PowerShell

Real-World Analogy



ROLE

Like a job title.
Example: **Accountant**



FEATURE

Like a skill or expertise.
Example: **Excel Expertise**



A job (Role) defines the main responsibility. Skills (Features) make the person more effective.

How Roles and Features Work Together



Windows Server



ROLE

Provides the main service (the what)



FEATURES

Add tools and capabilities (the how)



Better Services for Users

Roles deliver the service. Features make it better, more secure, and easier to manage.

Why It Matters



Helps you install only what you need—nothing more, nothing less.



Improves security by reducing unnecessary components.



Optimizes server performance and resource usage.



Makes management, maintenance, and troubleshooting easier.



Windows Services

Background services are programs that run on Windows systems to perform important tasks—without needing user interaction.



What is a Background Service?



A service is a program that runs in the background on Windows without user interaction. It starts automatically and continues to run even when no one is logged in.

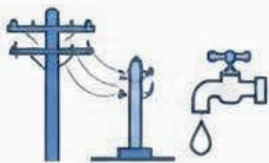


Why It Matters

- ✓ Services keep critical functions and applications running smoothly in the background.
- ✓ They enable servers and systems to run 24/7, handling tasks anytime, even without users being logged in.
- ✓ Essential for reliability, performance, and automation.



Real-World Analogy



Think of services like utilities in your home (electricity, water, gas). They work behind the scenes so you can focus on what you're doing—without needing to manage them all the time.

Services Work in the Background So You Can Work

BACKGROUND (SERVICES)

- DNS Service
- Print Spooler
- Windows Update
- Windows Update
- Other Services



WINDOWS SERVER

USERS (FRONTEND)

- Working on documents
- Browsing the web
- Using applications

Services run invisibly in the background, enabling everything to work.



Common Windows Services (Examples)



DNS Client (DNS Service)

Resolves domain names (like example.com) to IP addresses so you can access websites and network resources.



Print Spooler

Manages print jobs and queues documents to be printed, even when the printer is busy or offline.



Windows Update

Checks for, downloads, and installs updates to keep Windows secure and up to date.



Windows Defender Antivirus Service

Protects your system by running scans, monitoring threats, and staying active in the background.

What is Active Directory?



DEFINITION

Active Directory (AD) is a centralized system that manages users, computers, and permissions in a network.

It stores information about the objects in your environment and controls how they can access resources.



WHY IT MATTERS

- Centralizes management of users, computers, and resources.
- Ensures the right people have the right access.
- Improves security and reduces unauthorized access.
- Saves time with streamlined administration.
- Scales with your organization as you grow.



REAL-WORLD ANALOGY



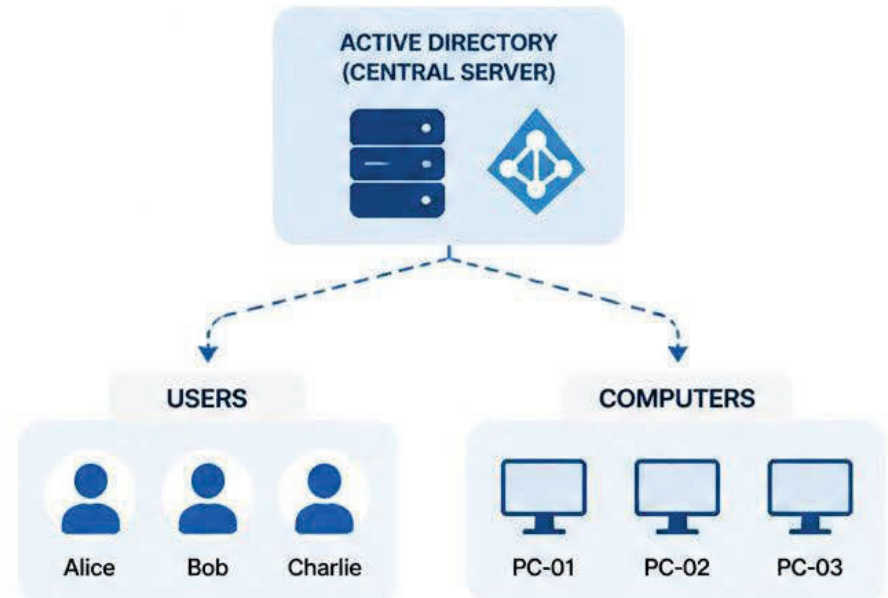
Active Directory is like a company's **HR department**.

It keeps track of all employees, their roles, access levels, and who can do what.



HOW IT WORKS

Active Directory uses **centralized management**. Administrators manage everything from one place, and the changes apply across the entire network.



Active Directory Domain Services (AD DS)



DEFINITION

Active Directory Domain Services (AD DS) is a Windows Server role that provides Active Directory functionality.



REAL-WORLD ANALOGY

AD DS is like the actual HR system software that runs the company's employee management. It stores employee records, enforces policies, and controls access to company resources.



WHAT IT DOES



Manages User Accounts

Creates, stores, and manages user accounts and their properties.



Manages Computer Accounts

Keeps track of computers and devices on the network.



Manages Security Policies

Applies and enforces security settings and policies across the organization.



Handles Authentication

Verifies users and computers when they sign in or access network resources.



WHY IT MATTERS

- Centralizes management of users, devices, and security.
- Improves security and consistency across the network.
- Makes it easier to manage access as your organization grows.
- Essential for most Windows-based organizations.

HOW AD DS WORKS



AD DS
on Windows Server



Users

Access to systems and applications



Computers

Managed and secured on the network



Shared Resources

Files, folders, and printers



Network Resources

Access controlled and monitored



Directory Services



DEFINITION

A directory service is a system that **stores and organizes information about network resources**, such as users, computers, and printers, making it easy to find and manage them.



WHAT IT STORES

A directory service can store many types of information, including:



Users

Names, usernames, passwords, email addresses, phone numbers, roles, and group memberships.



Computers

Computer names, IP addresses, operating systems, locations, and ownership.



Printers

Printer names, locations, IP addresses, and capabilities.



Other Resources

Shared folders, applications, policies, and more.



REAL-WORLD ANALOGY



A directory service is like a phonebook or company directory that lists everyone (people, departments, devices) and their contact information so you can quickly find what you need.



WHY IT MATTERS

Directory services provide a **central place** to look up and manage information. This ensures consistency, saves time, improves security, and makes it easier to manage resources across the network.





Domain Controller (DC)

The server that manages a domain.



Definition

A **Domain Controller (DC)** is a server running **Active Directory Domain Services (AD DS)** that manages a domain. It stores directory data and handles authentication, authorization, and other network services for users and computers.



What it does



Authenticates users

Verifies usernames and passwords when users sign in.



Stores user and computer info

Keeps information about users, computers, groups, and resources in the domain.



Enforces security policies

Applies rules for access, permissions, and other security settings across the domain.



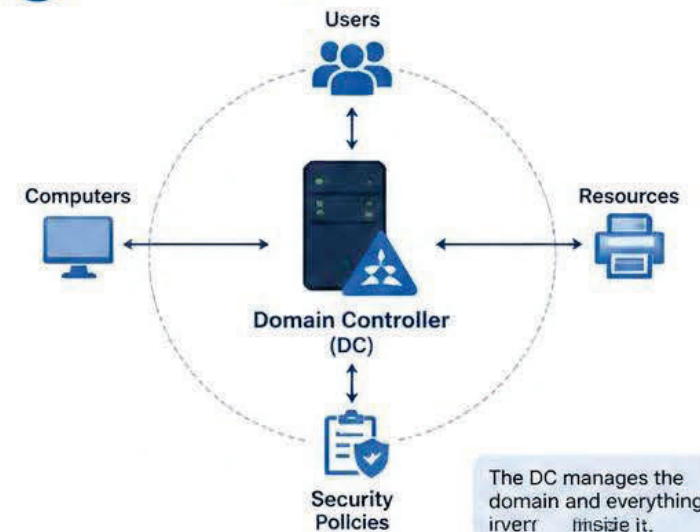
Real-world analogy



A Domain Controller is like a security guard at a company entrance. The guard checks IDs (authenticates users), keeps a log of who comes and goes (stores information), and enforces the company's rules (security policies).



How it fits together



Domain vs Domain Controller

	 A Domain	 A Domain Controller (DC)
 What it is	A logical grouping of users, computers, and resources that share a common database and security policies.	A physical or virtual server that runs AD DS and manages the domain.
 Type	A concept (logical)	A server (physical/virtual)
 Example	contoso.com	DC1.contoso.com
 Main role	Defines the boundary and scope for users, computers, and policies.	Enforces and manages that domain, handling authentication and directory services.

1.4

Object

Understanding what counts as an object in Active Directory (AD)



DEFINITION

An object is anything that is stored in Active Directory and can be managed.
Every object has properties (information) that describe it and help AD know how to handle it.



REAL-WORLD ANALOGY

Think of Active Directory like a filing cabinet. Each object is an item in the cabinet, and each item has its own folder with details about it.



TYPES OF OBJECTS



User Objects (People)

Represent users who sign in to the network.
Example: John Smith



Computer Objects (PCs/Servers)

Represent computers and servers joined to the domain.
Example: PC-101



Group Objects (Collections)

Represent collections of users, computers, or other objects.
Example: IT Support



Printer Objects

Represent printers available on the network.
Example: HP-OfficeJet



WHY IT MATTERS

Objects are the building blocks of Active Directory. Everything AD does—like giving access, applying policies, or managing resources—is done through objects.

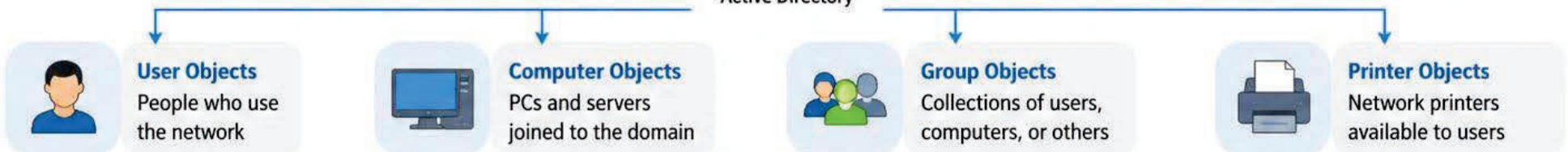
No objects = Nothing to manage.

More objects = More to organize and protect.

OBJECTS IN ACTIVE DIRECTORY



Active Directory





Attribute

Properties that describe an object.



What is an Attribute?

An attribute is a **property** or piece of **information** about an object.

Attributes describe the object and hold the actual data.



Real-World Analogy

Think of attributes like **fields on a form** or details in a **contact card**.



Why It Matters

Attributes store the **actual data** about an object. Without attributes, an object would have no details.

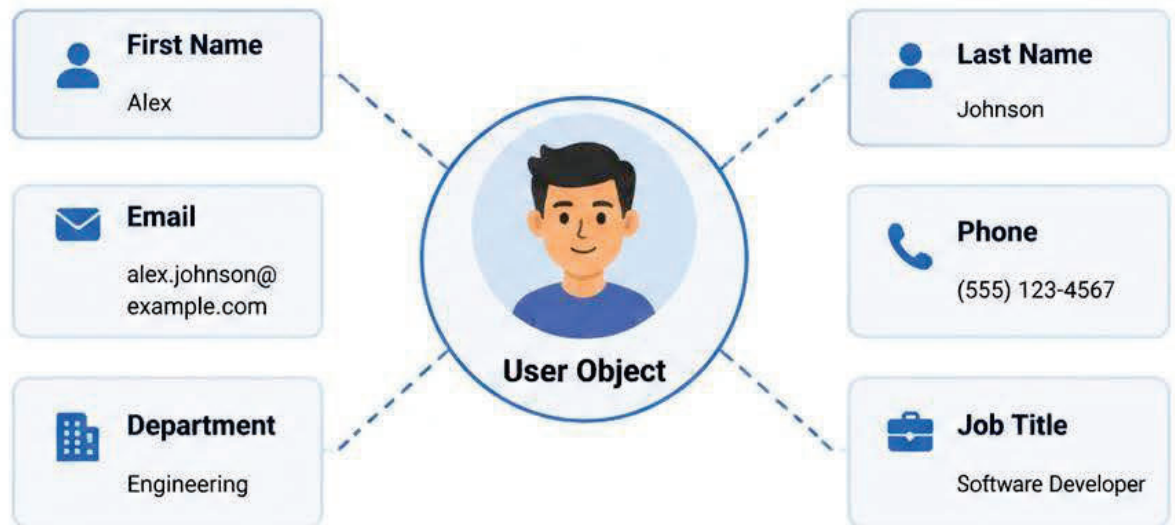


Example: User Object



User Object
(one user)

	First Name	Alex
	Last Name	Johnson
	Email	alex.johnson@example.com
	Phone	(555) 123-4567
	Department	Engineering
	Job Title	Software Developer





Schema

The set of rules that defines what objects can exist in AD and what attributes they can have.



WHAT IT CONTROLS

- **Defines object types** – what kinds of objects can exist in AD (e.g., User, Group, Computer).
- **Required vs. optional attributes** – which attributes are mandatory and which are optional for each object type.
- **Data types** – the type of data each attribute can store (e.g., String, Integer, Boolean, Date).



REAL-WORLD ANALOGY

Think of the schema like a form template.



Name *
Email *
Phone (optional)

This form says:

“Every User **must** have a **Name** and **Email**, and can optionally have a **Phone**.”



WHY IT MATTERS

The schema ensures consistency across the directory. By enforcing the same rules everywhere, it prevents invalid data, reduces errors, and enables applications and administrators to rely on a predictable structure.

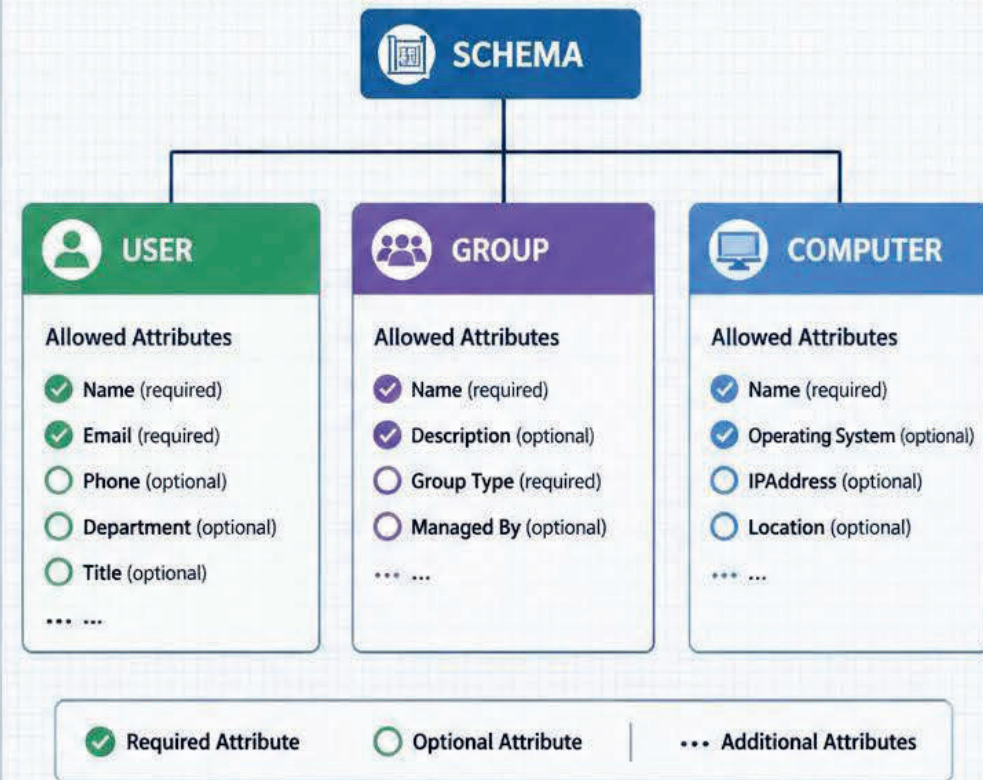


DEFINITION

The schema is the set of rules that defines what objects (such as Users, Groups, Computers) can exist in Active Directory and what attributes those objects can have.

SCHEMA AS A BLUEPRINT

Defines object types and the attributes each can have.



LDAP

Lightweight Directory Access Protocol



FOR BEGINNERS



LDAP (Lightweight Directory Access Protocol) is the language or method used to query and communicate with Active Directory (AD). It defines how applications send requests to AD and how AD returns the results.



WHAT IT DOES

LDAP allows applications to:



Search for objects

Find users, computers, groups, and more.



Retrieve information

Get details about objects (name, email, *memberOf*, location, etc.).



Modify objects

Create, update, or delete objects (such as user accounts or groups).



REAL-WORLD ANALOGY

Think of LDAP like a phonebook lookup system.



You ask:

"Find all users in Sales department."



The phonebook (AD) returns:

A list of matching people and their details.



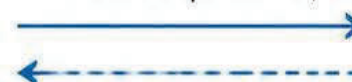
HOW IT WORKS (AT A HIGH LEVEL)



CLIENT

Application or tool

1. LDAP Query
(e.g., "Find all users in Sales department")



2. LDAP Response
(Results)



ACTIVE DIRECTORY (AD) SERVER

Holds directory data



WHY IT MATTERS

LDAP is the standard way to talk to Active Directory. Many tools and applications (like logon processes, HR systems, help desk tools, and admin utilities) use LDAP behind the scenes to find and manage information in AD securely and efficiently.



Kerberos

Authentication that's secure and convenient.

DEFINITION

Kerberos is an authentication protocol that verifies the identity of users and devices **without sending passwords over the network**.

HOW IT WORKS (SIMPLE)

1



User logs in

You enter your username and password once to the Kerberos system.

2



Receives a ticket

Kerberos verifies your identity and gives you a ticket.

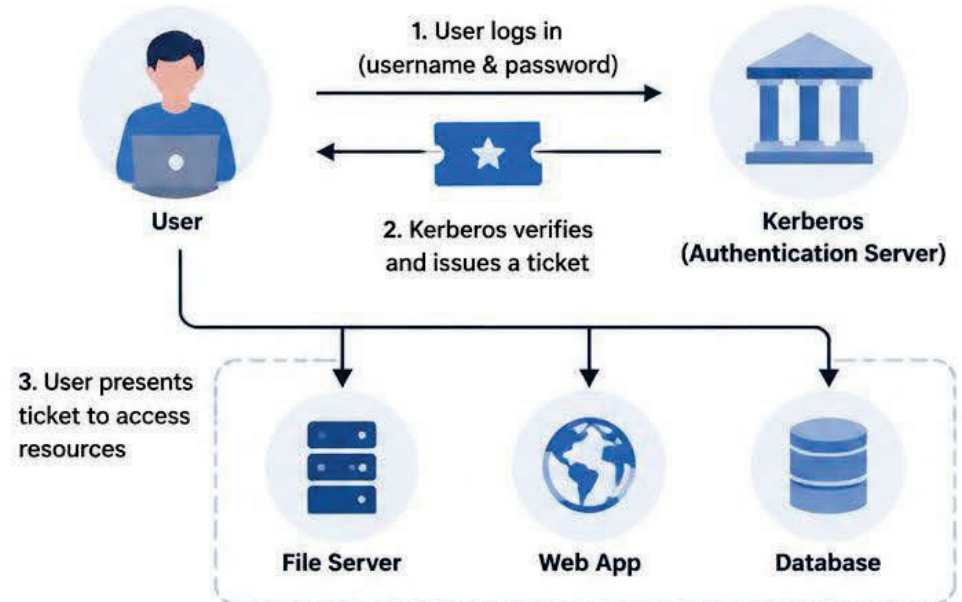
3



Uses ticket to access resources

You present the ticket to access different resources—no need to re-enter your password.

HOW IT WORKS (VISUAL)



The ticket proves your identity—no password needed again.

REAL-WORLD ANALOGY



Kerberos is like a concert ticket.

You prove your identity once at the entry, get a ticket, and then use that ticket to access different areas (seats, lounges, food court) without showing your ID again.

WHY IT MATTERS



More Secure

Your password is never sent over the network, reducing the risk of theft.



More Convenient

Log in once, access many resources without re-entering your password.

Global Catalog



DEFINITION

The Global Catalog is a special database that stores a copy of important information from **all** domains in a forest.



WHAT IT STORES



Partial copies of all objects

It stores a **partial set of attributes** for all objects from all domains in the forest.



Searchable index

It maintains an index of this information so it can be searched quickly.



WHY IT MATTERS



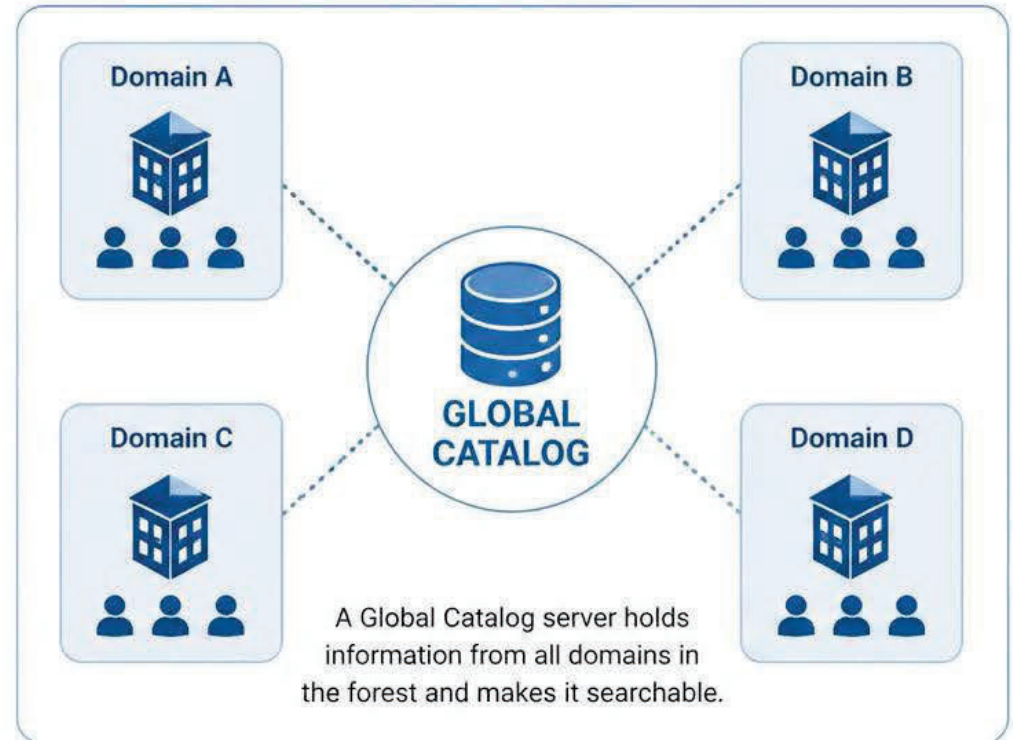
Enables fast searches across entire forest

Users and applications can find information from any domain **without knowing where it is located**.



Supports multi-domain environments

Essential for organizations with multiple domains that need to work together.



REAL-WORLD ANALOGY

The Global Catalog is like a **master index in a library** that lets you search all branches at once.





Forest



DEFINITION

A **Forest** is the highest level container in Active Directory that holds one or more trees.



WHAT IT CONTAINS



Multiple Trees

A forest can contain one or more trees.



Shared Schema

All objects in the forest use the same schema.



Shared Global Catalog

A single global catalog serves the entire forest.



Shared Configuration

Configuration data is shared across the forest.



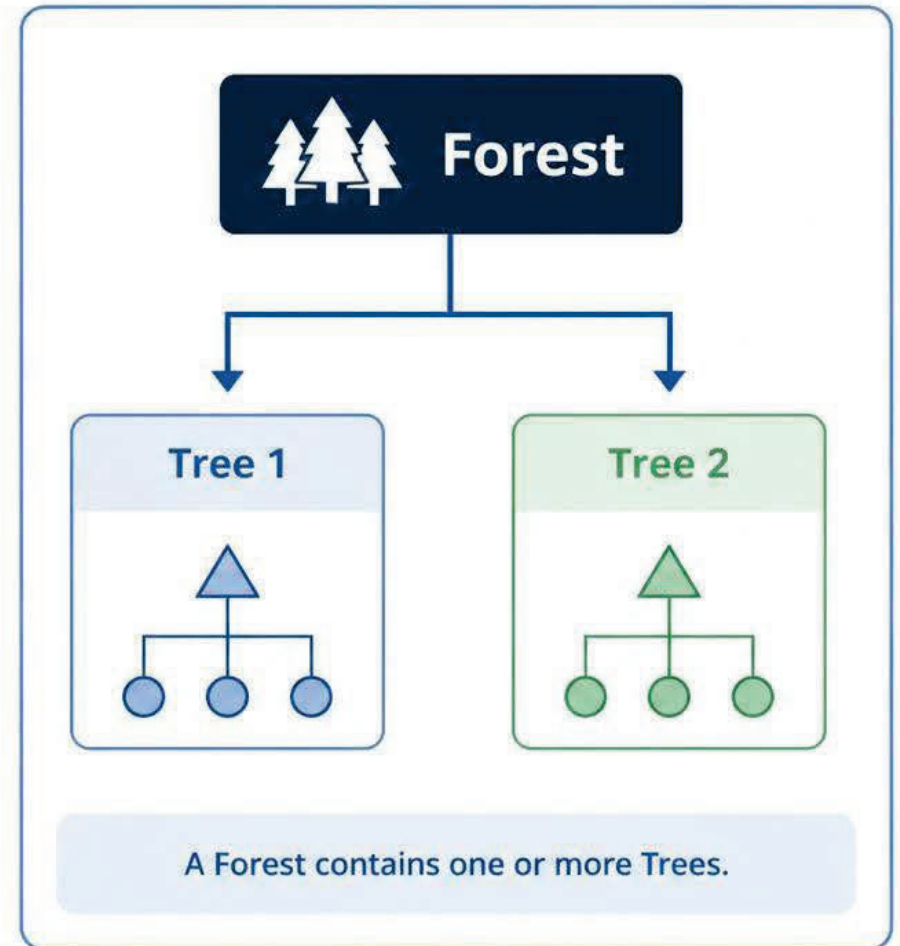
REAL-WORLD ANALOGY

A **Forest** is like a large corporation with multiple divisions (trees).



WHY IT MATTERS

Forests enable large organizations to have multiple domains (organized in trees) while sharing common infrastructure such as schema, global catalog, and configuration.





Tree

A collection of domains that share a contiguous namespace and trust relationship.



DEFINITION

A **Tree** is a collection of domains that share a contiguous namespace and trust relationship. Domains in a tree are organized hierarchically.



WHAT IT CONTAINS

A **Tree** contains multiple domains organized hierarchically under a common root domain.



RELATIONSHIP TO FOREST

Trees exist within Forests. A single forest can contain multiple trees.



REAL-WORLD ANALOGY

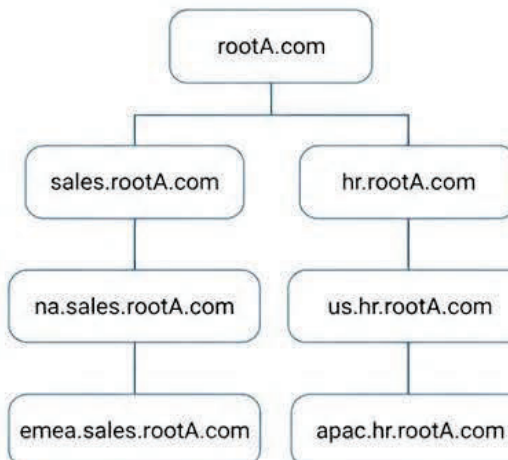
A **Tree** is like a division within a corporation. The division (tree) has multiple departments (domains) under a common structure and leadership.

TREES WITHIN A FOREST



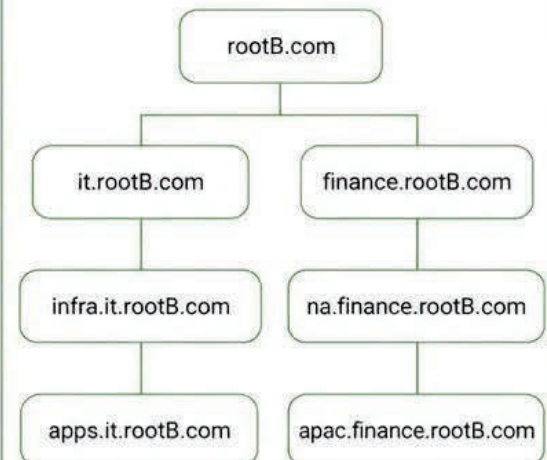
Contains multiple Trees

TREE A



Domains in Tree A share a contiguous namespace and trust relationship.

TREE B



Domains in Tree B share a contiguous namespace and trust relationship.



Domain



DEFINITION

A **Domain** is a logical grouping of computers, users, and resources that share a common namespace and security policy.



RELATIONSHIP TO TREE

- Domains are organized in Trees.
- Multiple domains can exist in a single tree, sharing a common structure.



REAL-WORLD ANALOGY

A Domain is like a department in a company.

Each department has its own manager (administrator), members (users and computers), and rules (policies).

WHAT IT CONTAINS



Users

People who can sign in and access resources.



Computers

Devices that are joined to the domain.



Groups

Collections of users and computers managed together.



Printers

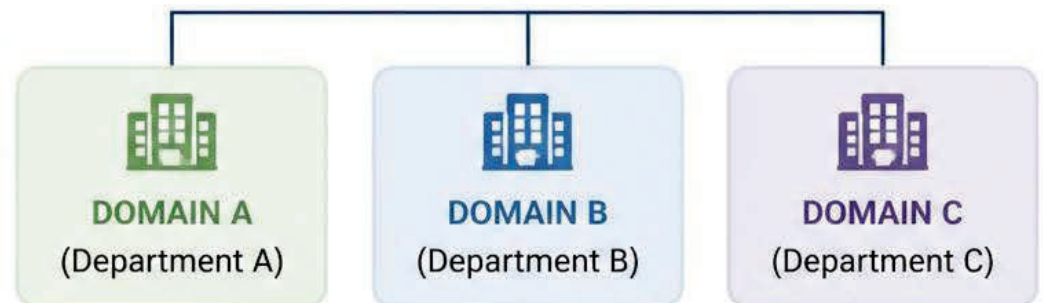
Shared printing resources.



Organizational Units (OUs)

Containers used to organize and delegate management.

TREE



One Tree can contain multiple Domains.



Organizational Unit (OU)

An Organizational Unit (OU) is a **container** within a domain used to organize users, computers, and other objects.



DEFINITION

An Organizational Unit (OU) is a container within a domain used to organize users, computers, and other objects.



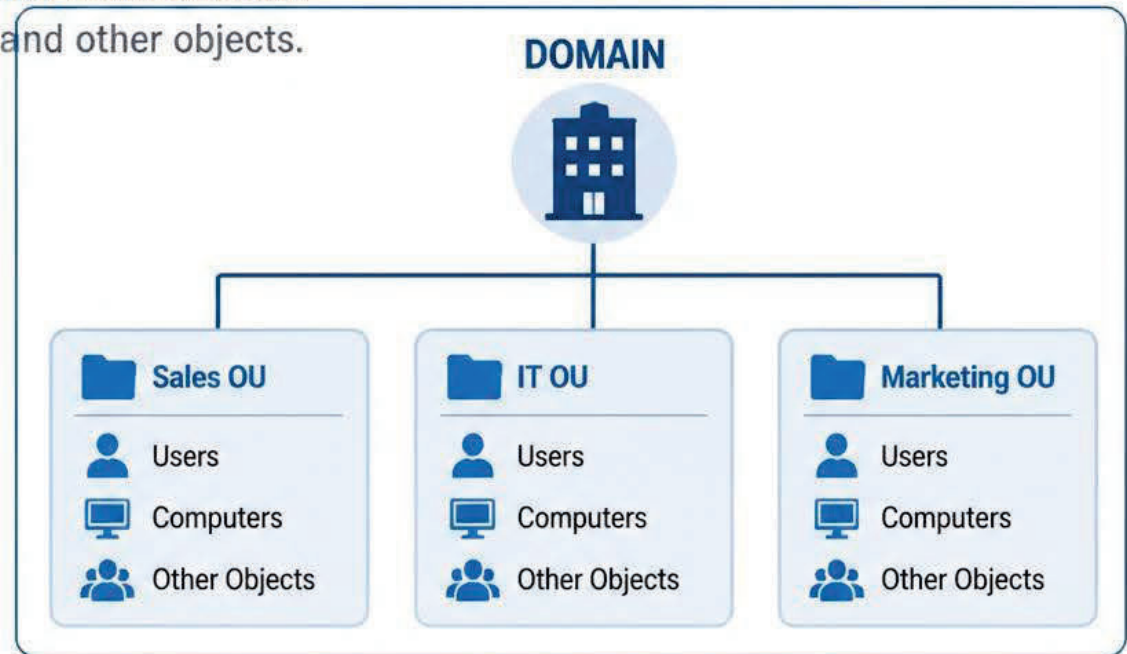
PURPOSE

- Organize objects by department, location, or function.
- Apply Group Policies to groups of objects.



REAL-WORLD ANALOGY

An OU is like folders in a filing cabinet. You create folders (OUs) to organize documents (objects) by category so they are easy to find and manage.



HOW IT'S USED



Sales OU

Contains all users and computers for the Sales department. Apply policies like desktop restrictions or software installation for Sales.



IT OU

Contains IT staff and servers. Apply policies for administrative access, security settings, and system management.



Marketing OU

Contains all Marketing users and computers. Apply policies for branding, software, and internet usage.



Sites



DEFINITION

Sites are logical groupings of computers and other network devices based on their physical network location.



PHYSICAL VS LOGICAL

Sites represent physical locations—such as offices or data centers—where network resources are located.

However, Sites are logical constructs defined in Active Directory (AD).



WHY IT MATTERS

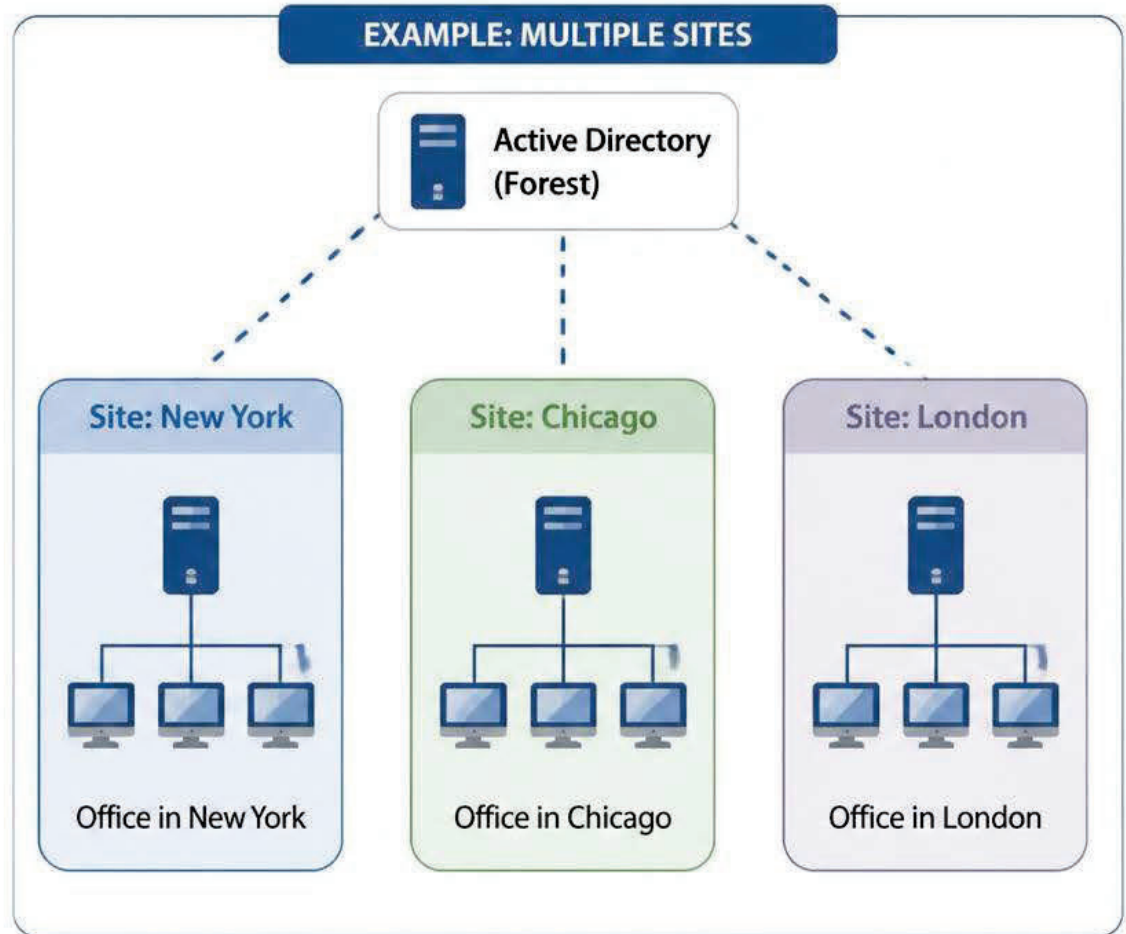
- Sites control replication traffic by determining how and when AD data is replicated between locations.
- Sites help improve authentication efficiency by directing clients to the closest domain controllers.



REAL-WORLD ANALOGY

Think of Sites like branch offices of a company located in different cities. Each office operates locally but stays connected to the others through corporate resources.

EXAMPLE: MULTIPLE SITES





AD Logs

What Gets Logged and Why It Matters



DEFINITION

AD Logs are records of events and changes that occur in Active Directory. They capture who did what, when, and where in the directory environment.

WHAT GETS LOGGED



User Logins

Records successful and failed logon attempts.



Password Changes

Logs when passwords are changed or reset.



Object Modifications

Tracks changes to users, groups, computers, and other AD objects.



Security Events

Records events like access denials, privilege use, and policy changes.



Replication

Logs replication events between domain controllers.

WHY IT MATTERS

- ✓ **Troubleshooting** – Helps identify the cause of issues.
- ✓ **Security Auditing** – Detects suspicious or unauthorized activity.
- ✓ **Compliance** – Supports regulatory and policy requirements.
- ✓ **Tracking Changes** – Provides an audit trail of who changed what and when.

REAL-WORLD ANALOGY



AD Logs are like a **security camera recording all activity** in a building. When something happens, you can review the footage to see what occurred and when.

AD LOGGING IN ACTION



Perform actions in Active Directory



AD
SERVER



Events are recorded in AD Logs

AD Database



DEFINITION

The AD Database is the **central storage system** that holds all Active Directory information.

WHAT IT STORES



User accounts



Computer accounts



Groups



Organizational Units



Security policies



Schema

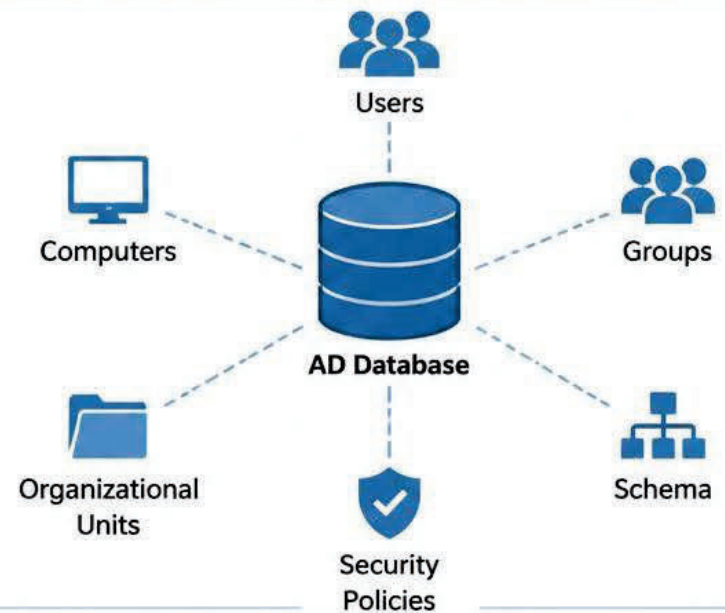
WHY IT MATTERS



All Active Directory data is stored here.

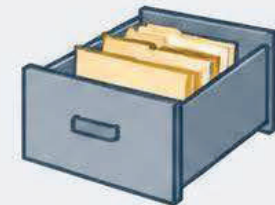
It must be **protected** and **backed up**.

AD DATABASE: CENTRAL STORAGE



REAL-WORLD ANALOGY

Think of the AD Database like a company's **master filing system**. It stores all important information in one place so everything can be found, managed, and kept secure.





NTDS.dit

THE HEART OF ACTIVE DIRECTORY



DEFINITION

NTDS.dit is the actual database file that stores all Active Directory data.



WHY IT MATTERS



This is the core of AD. It must be protected, backed up, and kept secure.



REAL-WORLD ANALOGY



NTDS.dit is like the actual **filing cabinet** containing all company records.



WHAT IT STORES



User accounts



Passwords (hashed)



Computer accounts



Groups



All AD objects and attributes



LOCATION



NTDS.dit is stored on **Domain Controllers**.



SYSVOL



DEFINITION

SYSVOL (System Volume) is a shared folder on Domain Controllers that stores Group Policy files and logon scripts.



WHAT IT STORES



Group Policy Objects (GPOs)

Contains the policy definitions and settings for GPOs.



Logon scripts

Scripts that run when a user logs on.



Startup scripts

Scripts that run when a computer starts.

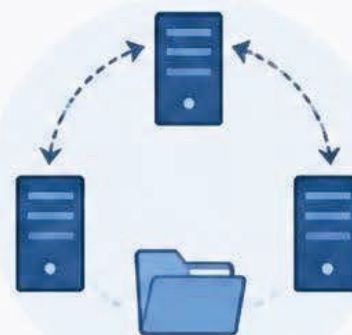


Domain-wide settings

Files and configurations that apply to the entire domain.



WHY IT MATTERS



SYSVOL is replicated across all Domain Controllers (DCs) using the Distributed File System Replication (DFSR). This ensures that all DCs have the same Group Policy files and scripts, providing consistent policies and settings across the domain.



LOCATION



Shared folder on each Domain Controller

C:\Windows\SYSVOL\sysvol\<domain>\



REAL-WORLD ANALOGY



Think of SYSVOL like a shared company policy manual that all managers have copies of. When the manual is updated, every manager receives the updated version so everyone follows the same rules.



Schema Configuration Partitioning

What this partition contains



DEFINITION

The Schema Configuration Partition is the part of the Active Directory database that stores schema definitions and configuration settings.



WHAT IT CONTAINS



Object class definitions

Defines the types of objects that can exist in AD.



Attribute definitions

Defines the properties of objects.



Replication settings

Defines how schema information is replicated.



Site topology

Contains site and service connection information.



WHY IT MATTERS

- Defines the structure and rules for the entire forest.
- Shared across all domains in the forest.

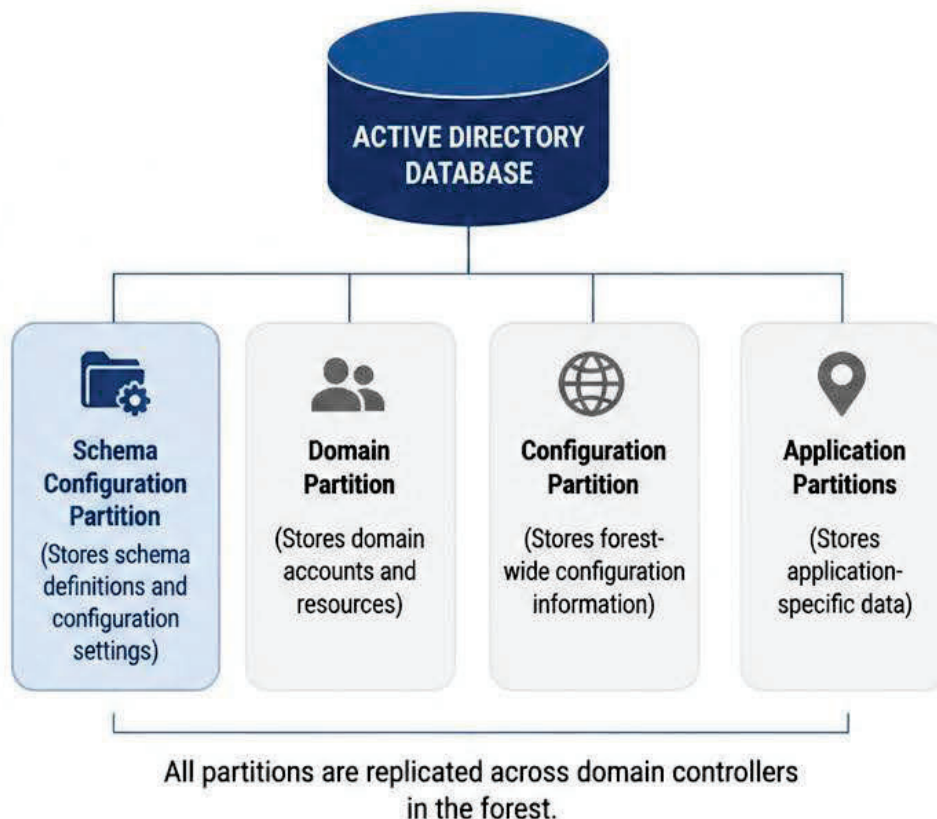


REAL-WORLD ANALOGY



The Schema Configuration Partition is like the master blueprint for the entire company—it defines the structure, standards, and connections used everywhere.

SCHEMA CONFIGURATION PARTITION AS PART OF AD DATABASE



Schema Domain Partitioning

What This Partition Contains



DEFINITION

A Schema Domain Partition is the part of the Active Directory (AD) database that stores **domain-specific** information.

WHAT IT CONTAINS



User accounts



Computer accounts



Groups



Organizational Units (OUs)



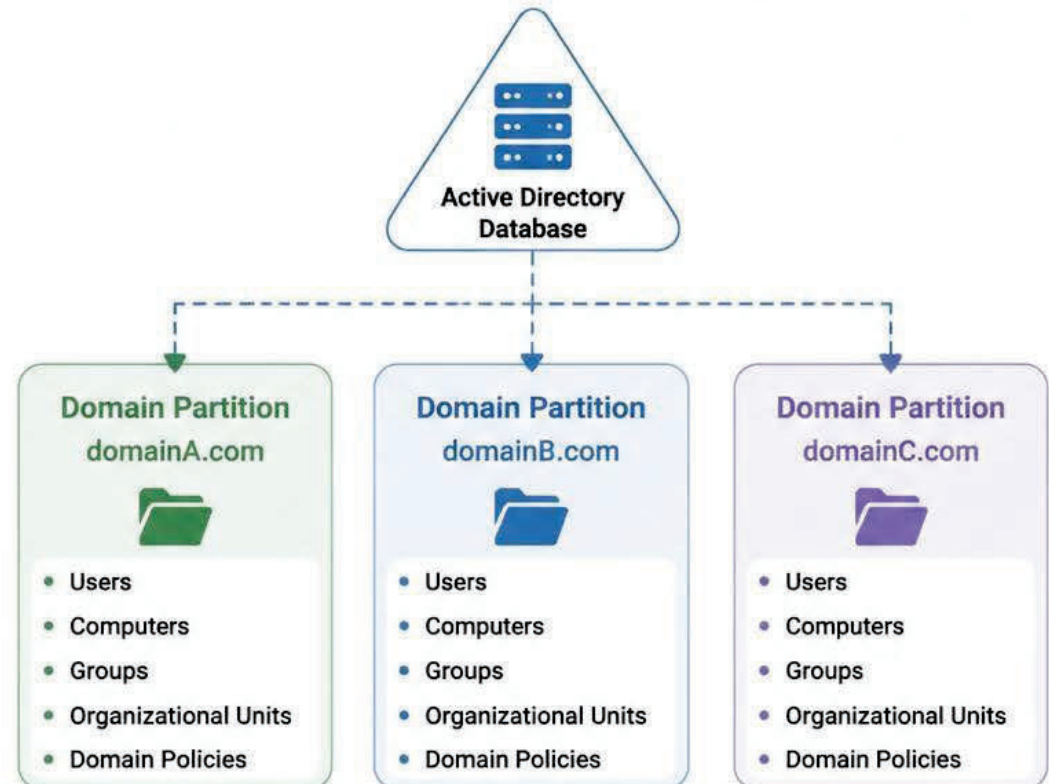
Domain-specific policies



WHY IT MATTERS

- Each domain has its own domain partition, containing its unique information.
- This allows multiple domains to coexist within the same Active Directory forest.

MULTIPLE DOMAIN PARTITIONS



REAL-WORLD ANALOGY

A Domain Partition is like a department's own filing system within a company. Each department (domain) keeps its own files (information) separate, but all are managed by the same company (Active Directory).



Schema Application Partitioning

Understanding what the Application Partition contains and why it exists.



DEFINITION

An **Application Partition** is an optional storage location within Active Directory that is used to store application-specific data.

WHAT IT CONTAINS



DNS Zones

Stores DNS zone data used by applications.



Application-Specific Objects

Includes objects created and managed by specific applications.



Custom Data

Stores custom or application-specific information.

WHY IT MATTERS



Allows applications to store their data in Active Directory without cluttering or overloading the main directory partitions.

REAL-WORLD ANALOGY



Think of an Application Partition as a special storage area in a workshop where specific tools and parts are kept separate from the general storage area.

WHEN USED



DNS

Stores DNS zone data in AD.



Exchange

Stores Exchange-specific configuration and data.



Custom Applications

Used by custom or third-party applications to store data.



STATIC IP

A fixed IP address that never changes.



DEFINITION

A Static IP is a fixed IP address assigned to a device that never changes. It remains the same until it is manually changed by the administrator.

DYNAMIC VS STATIC

DYNAMIC IP



Changes periodically (for client computers)

The IP address is assigned by the network and can change over time.

STATIC IP



Stays the same (for servers)

The IP address is fixed and does not change unless manually changed.



WHY SERVERS NEED STATIC IP

Clients need to know where to find the server. If the server's IP address changes, clients won't be able to connect to it.



REAL-WORLD ANALOGY



Static IP

Like a business address that doesn't change.



Dynamic IP

Like a temporary phone number that changes.

STATIC IP IN ACTION

SERVER



Static IP
203.0.113.10



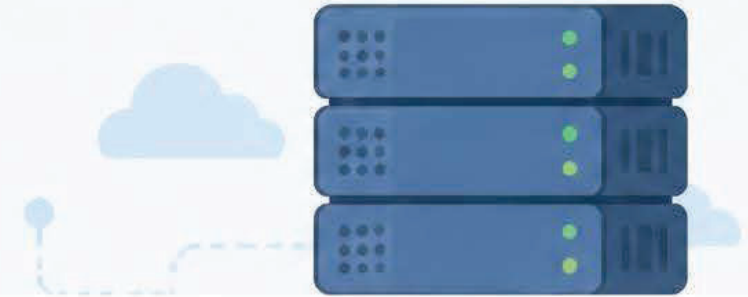
CLIENTS

Clients connect to the server using its fixed IP address.



Server Naming

Why naming a server matters.



DEFINITION

Server naming is the practice of assigning a meaningful name to a server so it can be easily identified.

WHY IT MATTERS



Easy Identification

Quickly recognize the server's purpose and location.



Helps with Management

Makes it easier to organize, monitor, and manage servers.



Appears in Logs and Network

Server names show up in logs, alerts, and across the network.



Makes Troubleshooting Easier

Clear names help identify issues faster and reduce confusion.

NAMING CONVENTIONS

Use clear, consistent names. Examples:



DC01

Domain Controller 1



DC02

Domain Controller 2



FILE-SERVER-01

File Server 1

REAL-WORLD ANALOGY



Calling employees
"Person 1, Person 2"
is confusing.



Naming employees
like "Alice (HR)" or
"Bob (IT)" is clear.

Server naming works the same way.



BEST PRACTICES

- Use descriptive names that reflect the server's role.
- Include the role and a number (if more than one server has the same role).



DNS Configuration

Why DNS must be set up before creating a domain



What is DNS?

DNS (Domain Name System) is the system that translates domain names (like example.com) to IP addresses (like 192.168.1.10) so computers can find and communicate with each other.



Why it's needed before AD

- Active Directory (AD) relies on DNS to locate domain controllers (DCs).
- Clients use DNS to find DCs and other domain resources by name.



Real-World Analogy

DNS is like a **phonebook**. You look up a name (domain name) and it gives you the phone number (IP address).



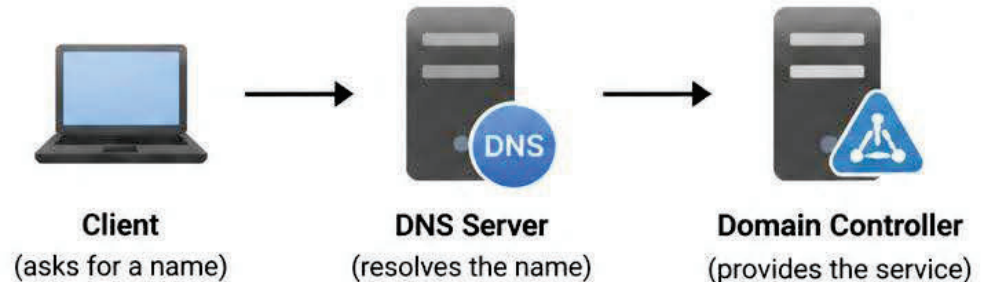
Setup

A **DNS server** must be installed and configured on the network before you promote a server to a Domain Controller. Without DNS, AD cannot function properly.



How it works

When you create a domain, AD automatically creates DNS records for domain services.



Query:
dc1.example.com



Response:
192.168.1.10

The client uses the IP address to connect to the DC.





Install AD DS Role

What happens during this step



DEFINITION

Installing the AD DS Role means adding the Active Directory Domain Services software to a Windows Server.



WHAT HAPPENS

- Windows installs AD DS binaries (files).
- Prepares the server for promotion to a Domain Controller.
- Adds the necessary AD DS services.



WHAT IT DOESN'T DO

- Doesn't create a domain yet.
- Doesn't make the server a Domain Controller yet.



REAL-WORLD ANALOGY

Installing AD DS is like installing tools before building something.



You have the tools on hand, but you haven't started building the structure yet.



NEXT STEP

After the installation is complete, you **promote** the server to a Domain Controller.

INSTALLATION PROCESS

1



Start Installation

You add the AD DS Role using Server Manager.



2



Install AD DS Binaries

Windows copies and installs AD DS files to the server.



3



Prepare Server

The server is configured and prepared for AD DS services.



4



Add AD DS Services

Necessary services are installed and configured (e.g., NTDS, DNS Integration, etc.).



Installation Complete

The AD DS Role is installed.
The server is ready to be promoted.



PROMOTE SERVER TO DOMAIN CONTROLLER



WHAT IS PROMOTION?

Promotion is the process of converting a regular Windows Server into a **Domain Controller**.

WHAT HAPPENS DURING PROMOTION?



1. Creates Active Directory (AD) database

A new AD database is created to store information about users, computers, groups, and other objects.



2. Initializes NTDS.dit

The AD database is stored in the **NTDS.dit** file, which becomes the core of the directory.



3. Sets up replication

The server is configured to replicate AD data with other domain controllers.



4. Configures DNS

DNS is configured to store AD-integrated zones and locate domain resources.



WHAT IT MEANS

After promotion, the server becomes a **Domain Controller (DC)**. It can:

- Authenticate users and computers
- Enforce security policies
- Manage domain resources
- Replicate directory data with other DCs



REAL-WORLD ANALOGY

Promotion is like promoting an employee from regular staff to manager. They get more responsibilities, authority, and the ability to manage others.



REQUIREMENTS

1

AD DS role must be installed first.

2

Static IP address must be configured.

3

DNS must be installed and ready.



Create New Forest

WHAT THIS ACTION DOES



DEFINITION

Creating a new forest is the process of establishing the top-level container for an entire Active Directory structure. It is the foundation on which all domains, trees, and objects are built.



WHAT IT DOES

- Creates the forest root domain
- Initializes the schema
- Sets up the Global Catalog
- Creates forest-wide settings



WHEN TO DO IT

- When setting up Active Directory for the first time
- When creating a completely separate Active Directory environment



REAL-WORLD ANALOGY

Creating a forest is like founding a new corporation. You establish the company itself, define the rules that apply across the entire organization, and set the foundation for all departments and operations.

FOREST CREATION – AT A GLANCE



1. Start Forest Creation



2. Initialize Forest Settings



3. Set Up Global Catalog



4. Forest Created



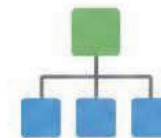
A new Active Directory forest is created and ready to host trees, domains, and objects.

WHAT IT CREATES



1. FOREST

The highest level container that contains all trees, domains, and objects.



2. FIRST TREE

The initial tree in the forest. A tree is a collection of domains that share a contiguous namespace.



3. FIRST DOMAIN

The root domain of the forest and the starting point for your directory structure.



Create Domain

What this action does



DEFINITION

Creating a domain means establishing a logical grouping within a forest or tree. It provides a boundary for managing users, computers, and resources with its own policies and administrators.



WHAT IT DOES

- ✓ Creates a domain container for objects such as users, computers, and groups.
- ✓ Sets up domain-specific policies and configurations.
- ✓ Creates a built-in Domain Administrators group for managing the domain.



WHEN TO DO IT

- After the forest exists.
- When adding new departments or locations that need their own administration and policies.



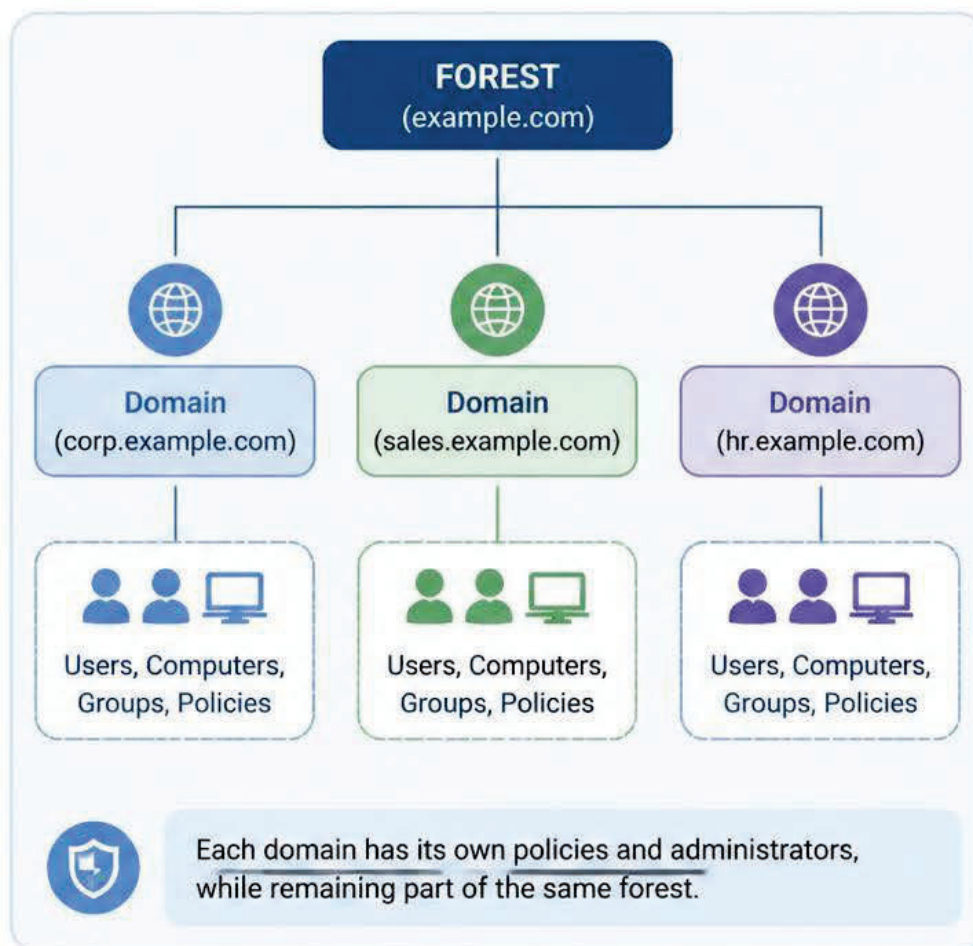
REAL-WORLD ANALOGY



Creating a domain is like opening a new branch office. It operates under the same company (forest) but has its own local management, rules, and staff.

RELATIONSHIP TO FOREST

Domains exist within forests (also called trees). A forest can contain one or more domains.





Domain/Forest Functional Levels

What this setting controls and why it matters



DEFINITION

Functional Levels are settings in Active Directory that determine which Windows Server versions can be Domain Controllers (DCs) in the domain or forest and what features are available.

WHAT IT CONTROLS



Minimum DC OS version

Sets the lowest Windows Server version that can be a DC in the domain or forest.



Available AD features

Determines which Active Directory features and capabilities are enabled.



Replication capabilities

Defines the level of replication and directory data functionality between DCs.

WHY IT MATTERS



Higher functional levels enable newer features and improvements.



But they require newer Windows Server versions to be DCs.



You can raise the functional level, but you cannot lower it.

EXAMPLES



2016 Functional Level

Supports Windows Server 2016 and newer DCs. Enables features up to the 2016 release.



2019 Functional Level

Supports Windows Server 2019 and newer DCs. Enables features introduced in 2019.



2022 Functional Level

Supports Windows Server 2022 and newer DCs. Enables the latest AD features and enhancements.



REAL-WORLD ANALOGY

Think of functional level like minimum employee skill requirements. If the requirement is set to "Senior," only employees with senior-level skills can join the team, and the team can use advanced tools and take on more complex projects.



CONFIGURE DNS ON DCs

Why Domain Controllers Typically Host DNS



DEFINITION

Configuring DNS on Domain Controllers (DCs) means installing and managing the Domain Name System service on DCs so they can resolve names and host the DNS records required for Active Directory and domain operations.



WHY DCs HOST DNS

- ✓ **AD Requires DNS:** Active Directory depends on DNS to locate domain controllers, services, and domain resources.
- ✓ **DCs Automatically Create DNS Records:** When a DC is promoted, it registers DNS records for itself and AD services automatically.
- ✓ **Centralized Management:** Hosting DNS on DCs keeps all directory-related records in one secure, reliable place.



WHAT DNS DOES FOR AD

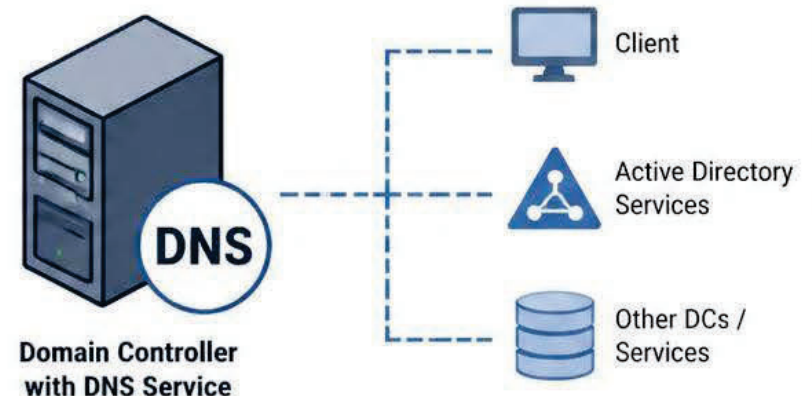
-  **Clients Find DCs Using DNS:** Clients query DNS to find domain controllers and logon services.
-  **Replication Uses DNS:** DCs use DNS to locate each other for replication and communication.
-  **Authentication Uses DNS:** Kerberos and other authentication processes rely on DNS to find the correct DC.



REAL-WORLD ANALOGY

DNS on a DC is like having the company phonebook at the main office. Everyone looks there to find the right person or department. If the main office has the phonebook, it's always up to date, trusted, and available to all branches.

DC WITH DNS SERVICE



HOW IT WORKS

When a server is promoted to a Domain Controller:

- 1 The DNS Server role is installed automatically (if not already present).
- 2 The DC registers its A record and other records (SRV, CNAME, etc.) in DNS automatically.
- 3 These records are replicated to other DCs, ensuring redundancy and high availability.



Active Directory Users and Computers (ADUC) Overview



DEFINITION

Active Directory Users and Computers (ADUC) is the primary management tool for Active Directory. It provides a graphical interface to manage users, computers, groups, organizational units (OUs), and permissions in your Active Directory environment.



WHAT IT DOES

- Create and delete users
- Create and delete computers
- Create and delete groups
- Manage organizational units (OUs)
- Set permissions and control access



WHY IT MATTERS

ADUC is where most daily AD management happens. From onboarding new users to organizing resources and securing access, ADUC is the go-to tool for keeping your directory organized and secure.



REAL-WORLD ANALOGY



Think of ADUC like the HR management system in a company, where you manage employees, their roles, teams, and access to company resources.

ADUC INTERFACE (CONCEPT)

Active Directory Users and Computers [DC01.contoso.com]		
Name	Type	Description
Alice Johnson	User	IT Specialist
Bob Smith	User	HR Manager
Marketing-PC01	Computer	Marketing Dept
Sales-PC02	Computer	Sales Dept
IT Support	Group	IT Support Team
HR Team	Group	Human Resources
Finance Team	Group	Finance Department

Left pane: Tree view of your Active Directory structure (domains, OUs, containers).

Right pane: Contents of the selected object (users, computers, groups, etc.).



HOW TO ACCESS

ADUC is a built-in Windows tool available on Domain Controllers and on admin workstations with the Remote Server Administration Tools (RSAT) installed.



Start Menu > Windows Administrative Tools > Active Directory Users and Computers



Create Users



DEFINITION

Creating a user means adding a new person to Active Directory as a user account. This account allows the person to sign in to the domain and use network resources.



PURPOSE

Each person in your organization needs a user account to:



Log in to the domain



Access shared resources and applications



Send and receive email



WHAT GETS CREATED



User account

The new person is added to Active Directory and can sign in to the network.



Email address (if configured)

An email address is created (e.g., john.doe@contoso.com).



Home folder (if configured)

A personal folder is created on the file server for the user to store their files.



REAL-WORLD ANALOGY

Creating a user is like hiring a new employee.

You add their name to the company records, give them an ID badge to get in (log in), provide the tools they need (resources), and set up their mailbox and workspace.



BASIC STEPS

1



Open ADUC

Open Active Directory Users and Computers (ADUC).

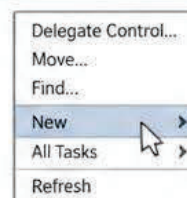
2



Right-click the OU

Right-click the Organizational Unit (OU) where you want to create the user.

3



Select New User

Point to **New** and click **User**.

4



Enter first name and last name

Type the user's first name and last name.

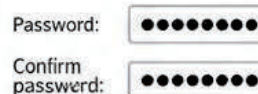
5



Create username

Enter a unique username (user logon name).

6



Set password

Enter and confirm a password for the user.



Click **OK** to create the user account.



Create Computers

Register a PC or server in Active Directory



DEFINITION

Creating a computer means registering a PC or server in Active Directory by creating a computer account in an Organizational Unit (OU).



PURPOSE

Each computer needs to be registered in Active Directory so it can:

- ✓ Join the domain
- ✓ Receive Group Policy and other settings
- ✓ Be managed and secured by IT



BASIC STEPS

- 1 Open ADUC**
Open Active Directory Users and Computers (ADUC).
- 2 Right-click the OU**
Navigate to the Organizational Unit (OU) where the computer account should be created. Right-click the OU.
- 3 Select New > Computer**
Choose **New**, then select **Computer**.
- 4 Enter the computer name**
Type the name of the computer you want to create (e.g., PC01).
- 5 (Optional) Pre-stage for security**
Pre-create the computer account before the device joins the domain to ensure only approved devices can be added.



WHAT IT DOES



Creates a computer account in Active Directory under the selected OU.



Allows the PC or server to join the domain using that account.



Enables the organization to apply policies, monitor, secure, and manage the device.

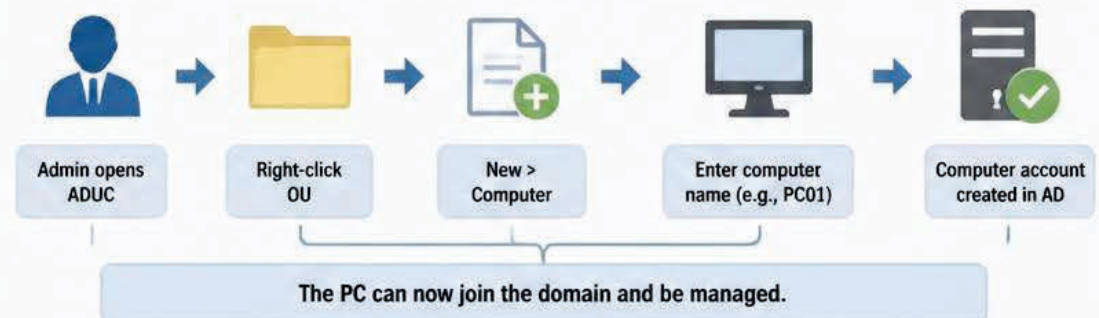


REAL-WORLD ANALOGY

Creating a computer account is like registering a company vehicle.

Before the vehicle can be used for business, it must be registered with the company so it can be tracked, insured, assigned policies, and properly managed.

COMPUTER CREATION – AT A GLANCE





Create Groups

Groups are collections of users that can be managed together.



DEFINITION

A group is a collection of user accounts that you can manage as a single unit. You can add users to a group, then assign permissions or access to the group instead of managing each user individually.



BASIC STEPS

-  **Open Active Directory Users and Computers (ADUC).**
From Server Manager, click [Tools > ADUC](#).
-  **Right-click the Organizational Unit (OU)**
where you want to create the group.
-  **Select New > Group.**
The New Object – Group window opens.
-  **Enter the group name and (optional) description.**
Example: [IT_Support](#), [HR_Team](#), [Sales_Users](#)
-  **Select the group scope.**
Choose one of the following:
[Domain Local](#) • [Global](#) • [Universal](#)



PURPOSE



Assign permissions to multiple users at once
Save time and ensure consistency.



Simplify management
Manage access in one place instead of for each user.



Organize by department or role
Keep users organized based on their function or responsibilities.



WHAT IT DOES

- ✓ Creates a group object in Active Directory.
- ✓ Allows you to add users to the group.
- ✓ Enables you to assign permissions to the group.

GROUP WITH MULTIPLE USERS



IT_Support
(Group)



User: Alice



User: Bob



User: Carol



User: David



REAL-WORLD ANALOGY

Groups are like teams or departments in an organization. Just as a department represents many people working together, a group represents multiple users that you can manage and give access to as a whole.



Assign PC to User

What it means in practice



DEFINITION

Assigning a PC to a user means linking a computer account to a user account in your organization.



WHAT IT MEANS

- The user can log in to that specific computer.
- The computer receives user-specific policies and settings.
- The user's files, preferences, and resources follow them.



HOW IT WORKS

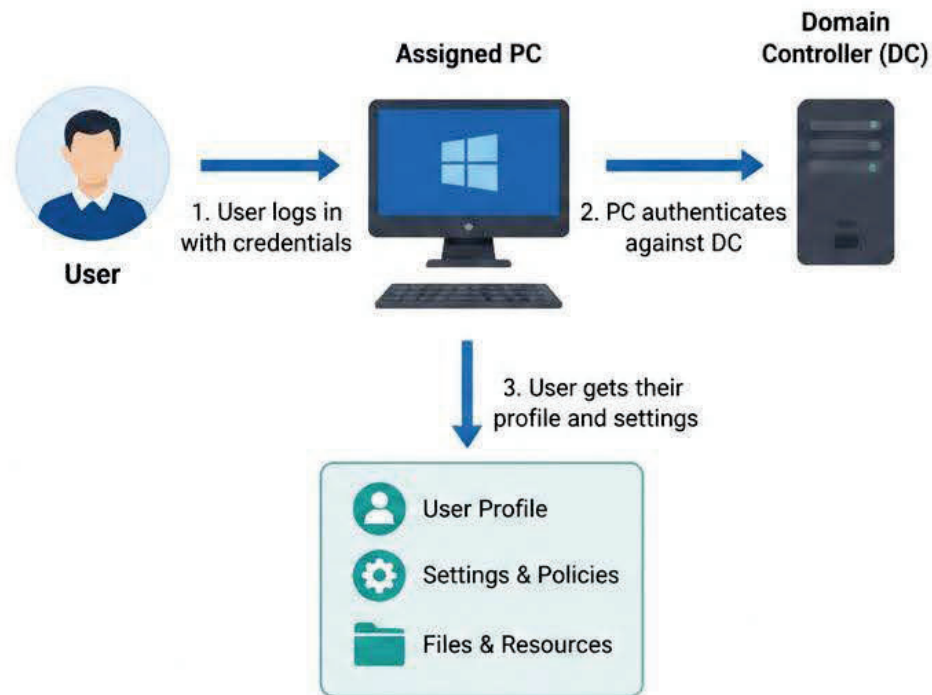
- The user logs in to the assigned PC with their credentials.
- The computer authenticates the user against the Domain Controller (DC).
- Once verified, the user gets their profile, settings, and access.



WHY IT MATTERS

- **Security:** Ensures only authorized users can use specific computers.
- **Personalization:** Users get their own experience every time they log in.
- **Management:** IT can apply the right policies, track usage, and provide better support.

HOW IT LOOKS



REAL-WORLD ANALOGY

Assigning a PC to a user is like assigning a desk to an employee. When they sit at that desk, they have access to their own tools, documents, and settings. Someone else sitting there won't have their things.



What's Next

Now that you have Active Directory set up, the next step is learning **Group Policy** — a powerful system that lets you manage settings across all computers and users in your domain automatically. You'll also explore permissions and security, learning how to control who can access what resources. Finally, you'll discover how to maintain and troubleshoot your AD environment to keep it running smoothly.



Great work getting AD DS up and running — you're building a strong foundation! Keep going; you're on the right path.